Retail Consumer Behavior Analysis

Project Report

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Tools: Python, Pandas, MySQL, Power BI

1. Project Overview

This project analyzes customer shopping behavior using a dataset containing 3,900 purchase records across multiple product categories. The objective is to identify patterns in spending, customer segments, product preferences, discount usage, and subscription behavior, and to translate the results into practical business recommendations.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Missing data: 37 values in the Review Rating column before cleaning.
- Customer information: age, gender, location, and subscription status.
- Purchase details: item purchased, category, purchase amount, season, size, and color.
- Shopping behavior: discount applied, promo code used, previous purchases, purchase frequency, review rating, and shipping type.
- **Dataset Source and Attribution**

  The dataset used in this project was obtained from Kaggle:

  Dataset Name: Shopping Trends and Customer Behaviour Dataset

  Author: Sahil Islam007

  Source: Kaggle Dataset

3. Exploratory Data Analysis Using Python

The dataset was loaded into Pandas for inspection, summary statistics, and data preparation. The following steps were documented in the project:

- Data loading: imported the dataset into a Pandas DataFrame.
- Initial exploration: used df.info() to inspect data types and structure, and df.describe() to calculate descriptive statistics.

In [5]:
```
df.describe(include = 'all')
```

Out[5]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN

Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900	3900.000000	3900	3900
6	2	2	NaN	6	7
Free Shipping	No	No	NaN	PayPal	Every 3 Months
675	2223	2223	NaN	677	584
NaN	NaN	NaN	25.351538	NaN	NaN
NaN	NaN	NaN	14.447125	NaN	NaN
NaN	NaN	NaN	1.000000	NaN	NaN
NaN	NaN	NaN	13.000000	NaN	NaN
NaN	NaN	NaN	25.000000	NaN	NaN
NaN	NaN	NaN	38.000000	NaN	NaN
NaN	NaN	NaN	50.000000	NaN	NaN

- Missing-value handling: imputed missing Review Rating values using the median rating for the corresponding product category.
- Column standardization: renamed columns to snake_case to improve readability and consistency.
- Feature engineering: created age_group by grouping customer ages and purchase_frequency_days from purchase-frequency information.
- Data consistency: checked whether discount_applied and promo_code_used conveyed redundant information and removed promo_code_used from the prepared data.

- Database integration: connected Python to MySQL and loaded the cleaned DataFrame into the customer_behavior database for SQL analysis.

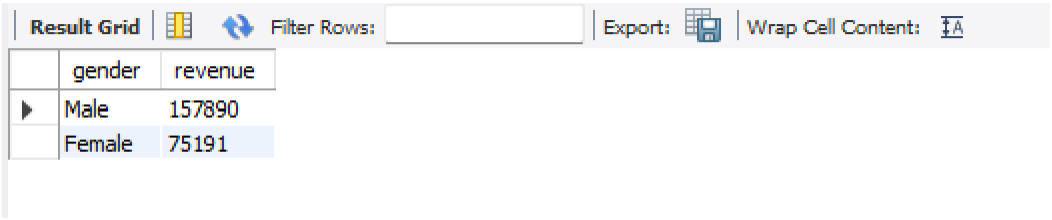
4. Data Analysis Using SQL

SQL queries were used to answer business questions and compare customer and product behavior. The following results are taken from the supplied project report.

Revenue by gender

Result: Female: 75,191; Male: 157,890.

Interpretation: The recorded revenue total is higher for male customers in this dataset.



gender	revenue
Male	157890
Female	75191

High-spending discount users

Result: 839 rows returned for customers who used discounts and spent above the average purchase amount.

Interpretation: Discount use can coexist with above-average purchase amounts; the result does not establish that discounts caused higher spending.

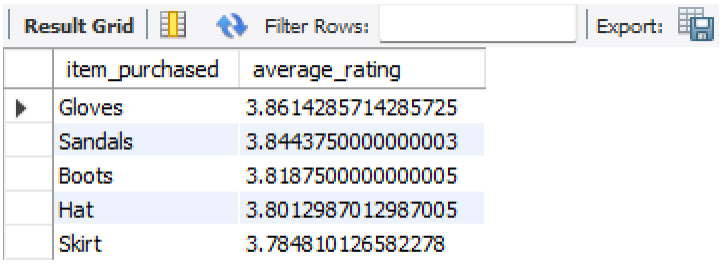
customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67
35	91
37	69
40	60
41	76
43	100
44	69

customer_id	purchase_amount
55	94
57	73
58	64
60	79
62	68
64	79
65	83
67	94
70	70
74	85
76	85
79	91
80	96
81	72
82	96
86	95
90	83
92	99
93	87
94	62

Top 5 products by average rating

Result: Gloves: 3.86; Sandals: 3.84; Boots: 3.82; Hat: 3.80; Skirt: 3.78.

Interpretation: These items have the highest average ratings among the products shown in the query output.



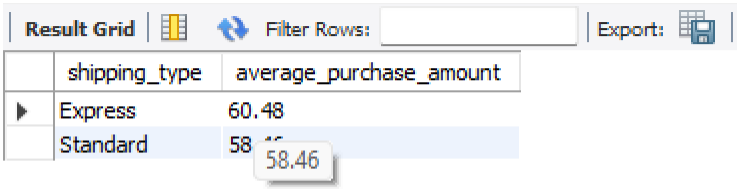
Result Grid | Filter Rows: | Export:

item_purchased	average_rating
Gloves	3.8614285714285725
Sandals	3.8443750000000003
Boots	3.8187500000000005
Hat	3.8012987012987005
Skirt	3.784810126582278

Average purchase amount by shipping type

Result: Standard: 58.46; Express: 60.48.

Interpretation: The recorded average is 2.02 higher for Express shipping.



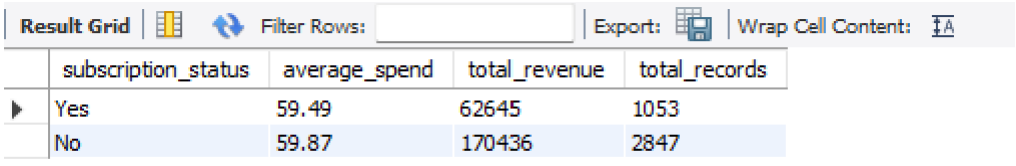
Result Grid | Filter Rows: | Export:

shipping_type	average_purchase_amount
Express	60.48
Standard	58

Subscribers vs. non-subscribers

Result: Subscribers: 1,053 customers; average spend 59.49; revenue 62,645. Non-subscribers: 2,847 customers; average spend 59.87; revenue 170,436.

Interpretation: Non-subscribers account for more total revenue in this dataset, while the average spend values are close.



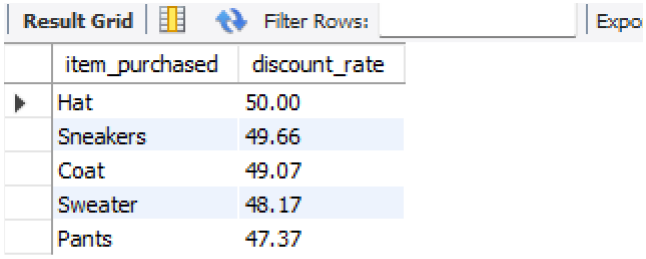
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

subscription_status	average_spend	total_revenue	total_records
Yes	59.49	62645	1053
No	59.87	170436	2847

Products with highest discount rates

Result: Hat: 50.00%; Sneakers: 49.66%; Coat: 49.07%; Sweater: 48.17%; Pants: 47.37%.

Interpretation: These items have the highest discount rates in the displayed result.



Result Grid | Filter Rows: | Expo

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

Customer segmentation

Result: Loyal: 3,476; Returning: 341; New: 83.

Interpretation: Most records fall into the Loyal segment, indicating that the majority of customers in this dataset have made more than five previous purchases. Returning customers have made between two and five previous purchases, while New customers have made zero or one previous purchase.

Result Grid | Filter Rows: | Export: | Wrap Cell

customer_segment	total_records
Loyal	3476
Returning	341
New	83

Top products per category

Result: Accessories: Jewelry (171), Sunglasses (161), Belt (161). Clothing: Blouse (171), Pants (171), Shirt (169). Footwear: Sandals (160), Shoes (150), Sneakers (145). Outerwear: Jacket (163), Coat (161).

Interpretation: The listed products have the highest order counts within their respective categories in the displayed query output.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

145

Repeat buyers and subscriptions

Result: Among customers with more than five purchases: subscription No = 2,518; Yes = 958.

Interpretation: Within the displayed repeat-buyer output, non-subscribers are more numerous.

Result Grid | Filter Rows: | Export:

subscription_status	repeat_buyers
Yes	958
No	2518

Revenue by age group

Result: Young Adult: 62,143; Middle-aged: 59,197; Adult: 55,978; Senior: 55,763.

Interpretation: Young Adults contribute the highest recorded revenue among the displayed age groups.

Result Grid | Filter Rows: | Export:

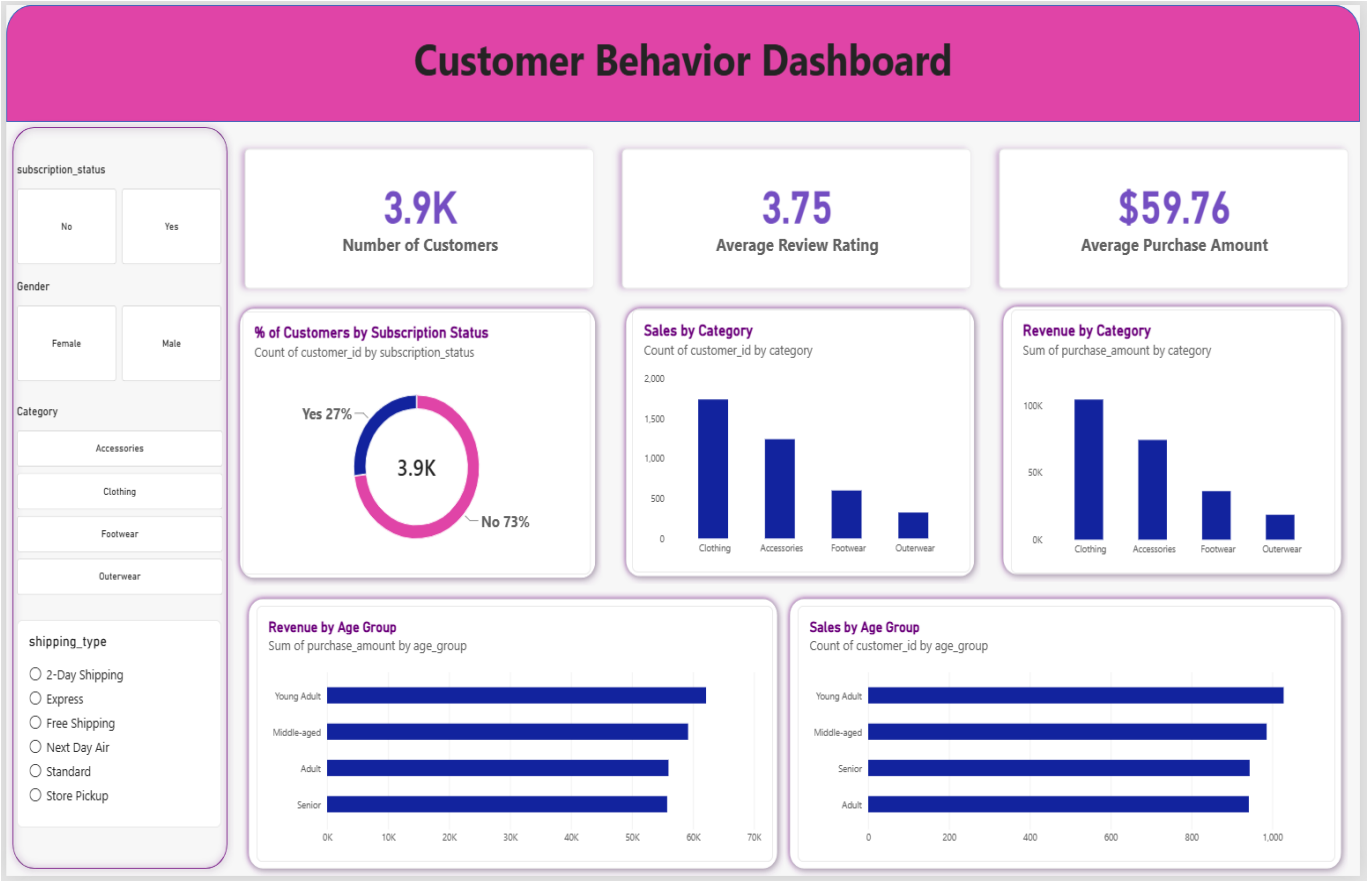
age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

55978

5. Dashboard in Power BI

An interactive Power BI dashboard was created to present the main indicators and comparisons visually. The supplied dashboard includes KPI cards for customer count, average purchase amount, and average review rating; a subscription-status donut chart; category-level revenue and sales visuals; age-group comparisons; and slicers for subscription status, gender, category, and shipping type.

- Number of customers: approximately 3.9K.
- Average purchase amount: $59.76.
- Average review rating: 3.75.
- Subscription split displayed: approximately 27% Yes and 73% No.



6. Business Recommendations

- Subscription growth: test subscriber-only benefits and measure whether they improve conversion and retention.
- Customer loyalty: consider rewards for repeat purchasers and evaluate movement between New, Returning, and Loyal segments.
- Discount policy: review discount-heavy products and compare sales lift with margin impact before changing promotions.
- Product positioning: feature highly rated and frequently purchased items in relevant campaigns.
- Targeted marketing: use age-group and category revenue patterns to guide experiments, while validating results over time.
- Shipping options: investigate the observed average-spend difference between Express and Standard shipping before making operational decisions.

7. Conclusion

The project demonstrates an end-to-end customer analytics workflow using Python and Pandas for data preparation, MySQL for structured business queries, and Power BI for visual reporting. The analysis describes differences in revenue by gender and age group, product ratings, discount rates, shipping-type averages, subscription status, and customer segments. These findings can help formulate business hypotheses and guide further investigation. Because the dataset is observational, the comparisons should be treated as descriptive rather than causal.